QUIZFORGE ACADEMIC ASSESSMENT

GATE COMPUTER SCIENCE - DATA STRUCTURES & ALGORITHMS

Part A: Algorithms and Asymptotic Analysis

1. What is the worst-case time complexity of searching for an element in an unbalanced Binary Search Tree with n nodes?
(A) O(1)
(B) O(log n)
(C) O(n)
(D) O(n log n)
Ans: (C)
Explanation: In an unbalanced skewed binary search tree (like a linked list), search degrades to O(n).

2. Which of the following sorting algorithms have an average time complexity of O(n log n)?
(A) Merge Sort
(B) Quick Sort
(C) Heap Sort
(D) Bubble Sort
Ans: (A)
Explanation: Merge sort, Quick sort, and Heap sort all achieve O(n log n) average time complexity.

3. What is the minimum number of queues needed to implement a Last-In First-Out (LIFO) Stack?
(A) 1
(B) 2
(C) 3
(D) 4
Ans: (B)
Explanation: Two FIFO queues are sufficient to simulate a LIFO stack behavior.

4. In a connected undirected planar graph with V vertices, E edges, and F faces, Euler's formula states that V - E + F equals:
(A) 0
(B) 1
(C) 2
(D) 4
Ans: (C)
Explanation: Euler's planar formula is V - E + F = 2.

5. A hash table of size 11 uses open addressing with linear probing and hash function h(k) = k mod 11. What index does key 25 occupy?
(A) 2
(B) 3
(C) 4
(D) 5
Ans: (B)
Explanation: 25 mod 11 = 3. Since slot 3 is initially empty, key 25 is placed at index 3.

ANSWER KEY & EXPLANATIONS

1. C
2. A
3. B
4. C
5. B